CHAPTER 8

SOLAR RADIATION HEAT BALANCE AND TEMPERATURE

CHAPTER ESSENCE AT A GLANCE:
Analyze solar insolation, spatial variability, atmospheric heating mechanisms (conduction, convection, advection), the Earth's global heat budget, factors governing temperature distribution, and temperature inversion.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Insolation and Solar Constant:** Insolation is incoming shortwave electromagnetic solar radiation; the solar constant at the top of the atmosphere averages approximately 1.94 calories per cm2 per minute (1361 W m^{-2}), varying with solar angle and daylight duration.

• **Atmospheric Heat Transfer Mechanisms:** Conduction heats air in direct contact with warm ground; Convection transports heat vertically via buoyant air currents; Advection transfers heat horizontally via winds (responsible for Loo winds in northern India).

• **Global Heat Budget Equilibrium:** Of 100 units of incoming solar radiation, 35 units are reflected directly back to space as planetary albedo; 14 units are absorbed by atmosphere and 51 units by Earth surface; surface reradiates 51 units as longwave terrestrial radiation, balancing the budget.

• **Controls of Temperature and Inversion:** Controlled by latitude, altitude, distance from sea, ocean currents, and air masses; Temperature inversion occurs when temperature increases with altitude during calm, clear, long winter nights.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$E = \sigma T^4 \rightarrow \text{Stefan Boltzmann Law of Total Terrestrial Radiant Flux}$

Radiative Emission Proportional to Fourth Power of Temperature

$\text{Earth Heat Budget (100 Units)} = \text{Albedo (35)} + \text{Atmosphere (14)} + \text{Surface (51)}$

Planetary Heat Energy Absorption and Reflection Partition

$I = I_0 \cos(\theta) \rightarrow \text{Lambert Cosine Insolation Reduction with Obliquity}$

Insolation Intensity Variation with Solar Zenith Angle

$\beta = \frac{Q_H}{Q_E} = \frac{\text{Sensible Heat Flux}}{\text{Latent Heat Flux}} \rightarrow \text{Bowen Energy Partitioning Ratio}$

Bowen Ratio Governing Sensible and Latent Boundary Fluxes

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The Earth's atmosphere is heated primarily from below by longwave terrestrial radiation (via conduction, convection, and advection), not directly by incoming shortwave solar insolation.

• **Alert:** Maximum insolation is received over subtropical hot deserts, not at the equator, because the equatorial belt experiences extensive cloud cover that reflects incoming solar rays.

• **Alert:** Temperature inversion in mountain valleys causes frost on valley floors, forcing fruit growers to plant orchards on higher valley slopes rather than valley bottoms.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Insolation. What is the value of the Solar Constant at the top of the atmosphere? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Insolation:** The incoming solar radiation received by the Earth's surface and atmosphere in the form of short electromagnetic waves. **[1 Mark]**

• **Solar Constant:** The solar energy received per unit area at the top of the atmosphere, averaging 1.94 calories per square centimetre per minute (or 1361 W m^{-2}). **[1 Mark]**

2. What is meant by the Albedo of the Earth? State its average value. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The fraction or percentage of incoming solar radiation reflected back into space by the Earth's surface and atmosphere without contributing to heating. **[1 Mark]**

• **Value:** The average albedo of the Earth is approximately 35 percent (35 units out of 100). **[1 Mark]**

3. Explain the three physical processes of heat transfer in the atmosphere (conduction, convection, advection). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Conduction:** Transfer of heat through molecular contact between the warm Earth surface and the immediately overlying layer of air. **[1 Mark]**

• **Convection:** Vertical upward movement of heated, expanded, less dense air, transferring heat to higher tropospheric altitudes. **[1 Mark]**

• **Advection:** Horizontal transfer of heat by moving air masses and winds, which is responsible for daily weather variations like hot 'Loo' winds. **[1 Mark]**

4. What is Temperature Inversion? State the ideal conditions necessary for its occurrence. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** A phenomenon in which temperature increases with increasing altitude instead of decreasing at the normal lapse rate. **[1 Mark]**

• **Ideal Conditions:** (1) Long winter nights allowing prolonged terrestrial radiative cooling, (2) Clear cloudless skies permitting unobstructed radiation escape, (3) Calm still air preventing turbulent mixing. **[2 Marks]**

5. Describe the Heat Budget of the Earth, accounting for incoming solar radiation and outgoing terrestrial radiation. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Incoming Shortwave (100 Units):** 35 units reflected to space as albedo (27 by clouds, 6 by air molecules, 2 by ice surfaces); 14 units absorbed by atmosphere; 51 units absorbed by Earth surface. **[2 Marks]**

• **Outgoing Terrestrial Longwave (51 Units):** Earth reradiates 51 units (17 units directly to space and 34 units absorbed by atmosphere). **[1 Mark]**

• **Atmospheric Radiative Balance:** Atmosphere returns 48 units (14 incoming + 34 terrestrial) back to space, ensuring net gain equals net loss and maintaining global thermal equilibrium. **[1 Mark]**

6. Analyze the major factors that control the spatial distribution of surface temperature over the globe.
[Score: 4 Marks]

Answer & Key Valuation Steps:

• **Latitude:** Angle of insolation decreases from equator to poles, causing temperatures to decrease poleward. **[1 Mark]**

• **Altitude:** High-altitude places are colder than plains due to the normal lapse rate (e.g. Ooty is cooler than Coimbatore). **[1 Mark]**

• **Distance from Sea (Continentality):** Water warms and cools slower than land; coastal areas experience moderate maritime climates, while interiors experience extreme continental extremes. **[1 Mark]**

• **Ocean Currents and Air Masses:** Warm currents raise coastal temperatures while cold currents lower temperatures and foster coastal desert fog. **[1 Mark]**